

MAXIMA MINIMA TRIGO

THEORY

Maximum value of $(2 \sin \theta + 3 \cos \theta)$ is

- (1) 2 (2) $\sqrt{13}$
(3) $\sqrt{15}$ (4) 1

(SSC Graduate Level Tier-I)

The maximum value of $\sin^4 \theta + \cos^4 \theta$ is

- (1) $\frac{1}{3}$ (2) 1
(3) 2 (4) 3

(SSC CGL Tier-I Exam, 09.08.2015
(1st Sitting) TF No. 1443088)

The minimum value of $2 \sin^2 \theta + 3 \cos^2 \theta$ is

- (1) 0 (2) 3
(3) 2 (4) 1

(SSC CPO (SI, ASI & Intelligence Officer) Exam 28.08.2011 (Paper-I))

The minimum value of $4 \tan^2 \theta + 9 \cot^2 \theta$ is equal to

- (1) 0 (2) 5
(3) 12 (4) 13

(SSC CHSL DEO & LDC Exam.
10.11.2013, Ist Sitting)

The minimum value of $\sin^2\theta + \cos^2\theta + \sec^2\theta + \operatorname{cosec}^2\theta + \tan^2\theta + \cot^2\theta$ is

- | | |
|-------|-------|
| (1) 1 | (2) 3 |
| (3) 5 | (4) 7 |

(SSC Graduate Level Tier-II
Exam. 16.09.2012)

Least value of
 $8\operatorname{cosec}^2x + 25\sin^2x$?

- 20√2
- 10√2
- 40√2
- 30√2

minimum value of $(25 \cos^2 x + 9 \sec^2 x)$?

SSC CGL 23/8/2021 (Morning)

(a) 30 (b) 20 (c) 15 (d) 40

The greatest value of $\sin x + \cos x$

NTPC 27/02/2021(Evening)

(a) $\frac{1}{4}$ (b) 1 (c) $\sqrt{2}$ (d) $\frac{1}{\sqrt{2}}$

$9\sin^2 \theta + 16 \cos^2 \theta$ का न्यूनतम मान क्या है?



(A) 0

(B) 9

(C) 16

(D) 25

$3\sin^2 \theta + 4 \cos^2 \theta$ का न्यूनतम मान है?



(A) 5

(B) 4

(C) 3

(D) 2

$2 - 2 \sin x - \sin^2 x$, $0 \leq x \leq \frac{\pi}{2}$ के सबसे बड़े मान से सबसे छोटे मान का अनुपात क्या है?

(A)-3

(B)-1

(C)1

(D)2

Minimum value of $4\tan^2 x + 9\cot^2 x$?

6

-6

12

-12

The minimum value of $\sec^2\theta + \operatorname{cosec}^2\theta$ is

- 0
- 2
- 1
- 4

The minimum value of $7\tan\theta + 8\cot\theta$ is

Options:

- A. $\sqrt{14}$
- B. $2\sqrt{14}$
- C. $3\sqrt{14}$
- D. $4\sqrt{14}$

The minimum value of $9 \cos^2 x + 2 \sec^2 x$ is

Options:

A. $\sqrt{18}$

B. $2\sqrt{18}$

C. $3\sqrt{18}$

D. $4\sqrt{18}$

The minimum value of $10 + \sec^2 x$ is

Options:

A. 10

B. 12

C. 11

D. 13

The sum of minimum and maximum value of $12\sin^2 x - 3$ is

Options:

A. 3

B. 6

C. 9

D. 12

The sum of minimum and maximum value of $11 + \cos^2 x$ is

Options:

A. 11

B. 12

C. 23

D. 24

The minimum value of $10\tan^2 x + 15\sec^2 x$ is

Options:

A. 12

B. 13

C. 14

D. 15

$\cos^3 \theta + \sec^3 \theta$ where $0^\circ \leq \theta \leq 90^\circ$ का न्यूनतम मान क्या है?

A) 0

B) 2

C) 1

D) None of above

$$\sin^{29}\theta.\cos^{29}\theta$$

$$\sin^{140}\theta.\cos^{140}\theta$$

Minimum and maximum value of $\cos^2 x + \sec^2 x$?

Where $0 \leq x \leq \pi/2$

0,0.5

0.5,1

1,2

$y \geq 2$

Minimum value of $\sin^4 x + \cos^4 x - 2\sin^2 x \cos^2 x$?

0

2

1

Doesn't exist

MAXIMUM,
 $1+2\sin^2x\cos^2x-\sin^4x-\cos^4x$,
 $0<x<1$?

1

2

3

4

Maximum and minimum
value of
 $5\sin^2x+10\cos^2x+12\sin x\cos x$?

1,12

0,14

1,14

-1,14

Maximum value of
 $\sin^8 x + \cos^{14} x$?

0

1

2

3

Minimum and maximum value of
 $\sin^6 x + \cos^6 x$?

$1/4, 1$

$-1/4, 1$

$0, 1$

$0, 1/4$

Minimum and maximum value of
 $\sin^4 x + \cos^4 x$?

$1/2, 1$

$-1/2, 1$

$0, 1$

$0, 1/2$

Minimum and maximum value of
 $\sin^2 x + \cos^4 x$?

$1/2, 1$

$3/4, 1$

$-3/4, 1$

$0, 1/2$

Minimum and
maximum value of
 $\sin^4 x \cdot \cos^4 x$

Thankyou